

Design Proposal: Mach-Zehnder Interferometers for Silicon Waveguide Group Index Extraction

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Abstract—This proposal describes three unbalanced Mach-Zehnder interferometers (MZIs) built from 500 nm × 220 nm silicon strip waveguides for quasi-TE operation at 1550 nm. The waveguide mode was simulated with Lumerical MODE, and a second-order compact model of the effective index was extracted. Using this model, the free spectral range (FSR) of each MZI was predicted and the circuits were simulated in Lumerical INTERCONNECT. Path length differences of 65, 145 and 745 μm give FSRs of 8.8, 3.9 and 0.77 nm, all within the ~50 nm measurement bandwidth, so that the group index can be extracted from measured spectra.

Index Terms—Mach-Zehnder interferometer, silicon photonics, group index, compact model, free spectral range.

I. INTRODUCTION

Silicon photonics uses the high index contrast between silicon and its oxide cladding to build compact, CMOS-compatible optical circuits for communications, sensing and computing. Accurate knowledge of the waveguide group index, n_g , is essential for designing interferometers and resonators, because it sets the spacing of spectral features and the delay of optical signals.

The Mach-Zehnder interferometer is a simple and robust structure for measuring n_g . Light is split into two arms of different lengths and recombined, producing a periodic transmission spectrum whose FSR depends directly on n_g and on the path length difference ΔL . The design objective of this work is to build a set of MZIs with different ΔL from which the group index can be extracted experimentally and compared with simulation. Although MZIs are well established in photonics, studying them in depth in this course will help me better understand the underlying concepts.

II. THEORY

An input field E_i is split equally by a Y-branch, so each arm carries $E_i/\sqrt{2}$. After propagating lengths L_1 and L_2 with propagation constant $\beta = 2\pi n_{\text{eff}}/\lambda$ (neglecting loss), the two arms acquire phases βL_1 and βL_2 . They are recombined by a 50/50 directional coupler, which gives two complementary outputs. Neglecting the constant phase of the coupler, the field at one output port is

$$E_o = (E_i/2)[e^{-j\beta L_1} + e^{-j\beta L_2}] \quad (1)$$

The output intensity is therefore

$$I_o = (I_i/2)[1 + \cos(\beta\Delta L)] \quad (2)$$

and the other port gives $(I_i/2)[1 - \cos(\beta\Delta L)]$, where $\Delta L = L_2 - L_1$. Transmission maxima occur where $\beta\Delta L = 2\pi m$. Expanding β in wavelength gives the spacing between adjacent maxima, the free spectral range:

$$\text{FSR} = \lambda^2 / (n_g \Delta L), \quad n_g = n_{\text{eff}} - \lambda \, dn_{\text{eff}}/d\lambda \quad (3)$$

Measuring the FSR of a device with known ΔL thus yields n_g directly. Using several ΔL values allows the result to be checked and reduces the effect of fabrication variation.

III. MODELLING AND SIMULATION

A. Waveguide

The waveguide is a silicon strip, 500 nm wide and 220 nm high, on a buried oxide with oxide cladding, designed for the quasi-TE mode at 1550 nm. The eigenmodes were computed with the finite-difference eigenmode (FDE) solver in Lumerical MODE. The fundamental mode at 1550 nm has $n_{\text{eff}} = 2.4473$, $n_g = 4.2031$ and a TE polarization fraction of 98%. Its electric field intensity is shown in Fig. 1.

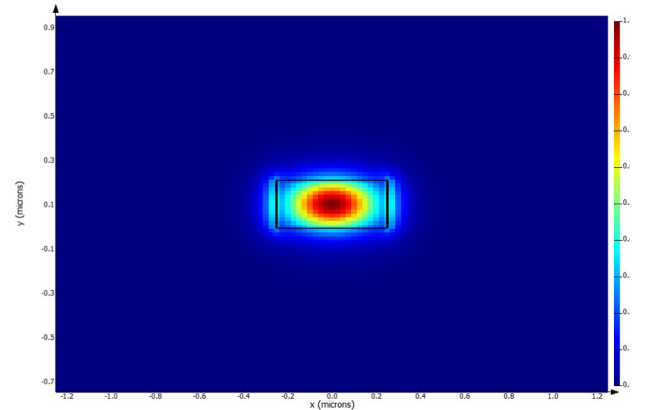


Fig. 1. Electric field intensity of the fundamental quasi-TE mode of the 500 nm × 220 nm strip waveguide at $\lambda = 1550$ nm (Lumerical MODE).

A frequency sweep from 1550 to 1600 nm gives the effective and group index curves shown in Fig. 2. The effective index decreases with wavelength, while the group index rises slightly from 4.203 to about 4.210.

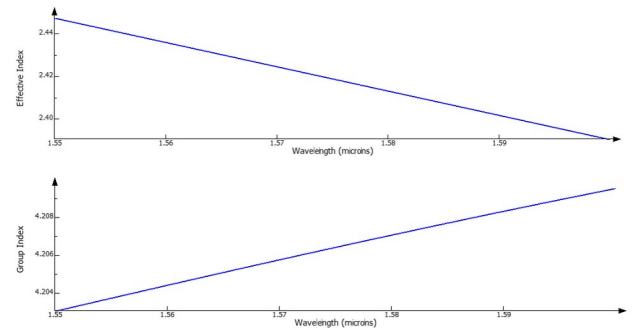


Fig. 2. Effective index (top) and group index (bottom) of the fundamental TE mode versus wavelength, from the Lumerical MODE frequency sweep.

B. Compact Model

A second-order Taylor expansion of n_{eff} around $\lambda_0 = 1.55 \mu\text{m}$ was fitted to the MODE results (n_{eff} and n_g at 1.50 and 1.55 μm):

$$n_{\text{eff}}(\lambda) = 2.4473 - 1.1327(\lambda - \lambda_0) - 0.0459(\lambda - \lambda_0)^2 \quad (4)$$

with λ in μm . The model agrees with the MODE frequency sweep in Fig. 2 (e.g., $n_{\text{eff}} = 2.3906$ at $1.60 \mu\text{m}$).

C. Interferometer Design

Three MZIs were designed using the default components of the SiEPIC EBeam PDK: TE fiber grating couplers, a Y-branch splitter, a broadband directional coupler as the combiner, and 500 nm strip waveguides. The arm lengths were set with the SiEPIC waveguide length tool; each short arm is approximately $35 \mu\text{m}$. Each circuit has three grating couplers on a $127 \mu\text{m}$ pitch, one for the input and two for the complementary outputs of the directional coupler. Fig. 3 shows the layout. The expected FSR of each device, calculated from (3) with $n_g = 4.203$, is listed in Table I. All FSRs are well below the $\sim 50 \text{ nm}$ measurement bandwidth, so several fringes will be visible in every spectrum, and the wide range of ΔL lets the FSR-versus- ΔL trend be checked.

TABLE I
MZI Design Parameters and Expected FSR

Device	Long arm (μm)	ΔL (μm)	FSR @1550 (nm)
MZI 1	100	65	8.79
MZI 2	180	145	3.94
MZI 3	780	745	0.77

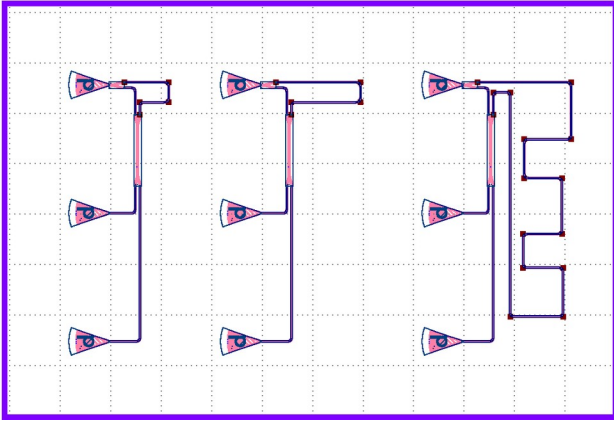


Fig. 3. Layout of the three MZIs (KLayout, SiEPIC EBeam PDK), with ΔL increasing from left to right. MZI 3 uses a folded long arm to fit the $780 \mu\text{m}$ path in the allotted area.

D. Circuit Simulation

The full circuits, including grating couplers, Y-branch and directional coupler, were simulated in Lumerical INTERCONNECT (Fig. 4). The grating couplers impose the roughly parabolic envelope with a peak insertion loss of about 5 dB near 1550 nm , and the two output ports are complementary. The simulated FSRs are about 8.8 nm for MZI 1, about 4 nm for MZI 2 and about 0.72 nm for MZI 3 (bottom panel of Fig. 4), consistent with Table I given the approximate short-arm length.

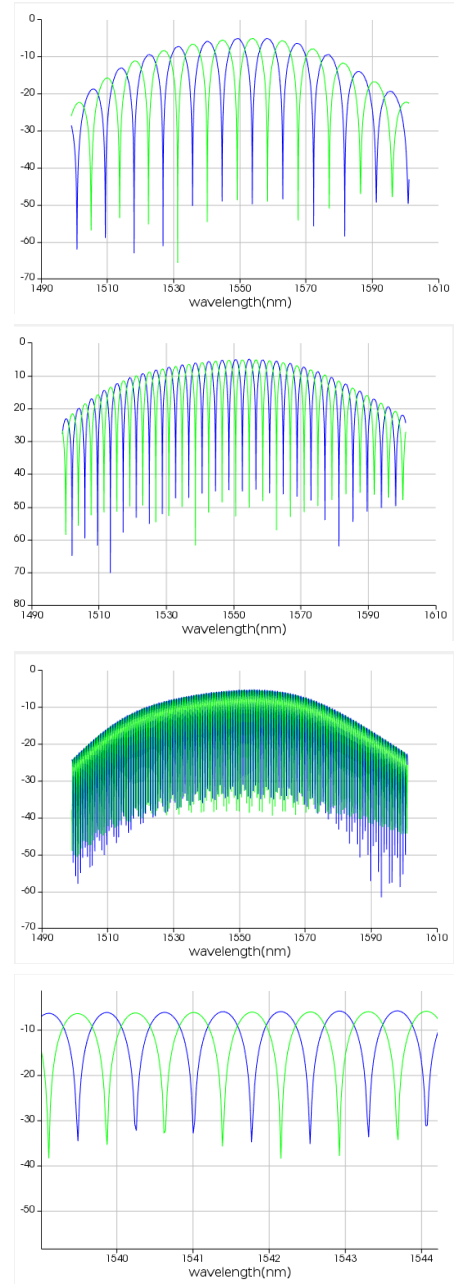


Fig. 4. INTERCONNECT transmission spectra (TE gain, dB) including grating couplers. From top to bottom: MZI 1, MZI 2, MZI 3, and MZI 3 zoomed to $1539\text{--}1544 \text{ nm}$ (FSR $\approx 0.72 \text{ nm}$). Blue and green traces are the two output ports.

IV. FABRICATION

To be completed after layout submission.

V. EXPERIMENTAL DATA

To be completed after measurement.

VI. ANALYSIS

To be completed: the measured FSR versus ΔL will be used to extract n_g and compare it with the simulated value of 4.20.

VII. CONCLUSION

To be completed.

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